C++ PROJECT PRESENTATION

Software-Based Ray Tracer

High-performance 3D rendering engine featuring global illumination, BVH spatial indexing, and multi-core parallelization.

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CORE CHALLENGE

Problem Statement

Design and implement a standalone, high-performance C++ software ray tracer capable of rendering complex 3D scenes with global illumination effects, optimized using spatial acceleration structures (**BVH**) and multi-core thread parallelization.

3D Geometry
Complex Primitives

High Speed
BVH + Multi-Core

TECH STACK

Technology Used

C++20 Standard

Leverages modern language features, concepts, and zero-cost memory management abstractions.

OpenMP Threads

Parallelizes scanline rendering across multi-core CPU architectures for linear speedups.

Eigen Library

High-performance linear algebra template library for matrix and vector calculations.

CAPABILITIES

Main Features of Application

Global Illumination

Monte Carlo path tracing simulates realistic light bounces, color bleeding, and soft area shadows.

Spatial Acceleration

BVH spatial tree indexing reduces ray intersection checks from linear O(N) to logarithmic O(log N).

Advanced Materials

Fully supports Lambertian surfaces, metallic gloss reflections, and dielectric glass refraction.

ARCHITECTURE • MODULE 01

Math & Geometry Core

Vector & Ray Primitives

Implements fundamental 3D vector arithmetic, ray equations, and intersection tests for spheres, planes, and polygon meshes.

`Inline optimized for raw speed`

Camera & Viewport

Configurable pinhole camera supporting field-of-view angles, depth-of-field lens apertures, and anti-aliased primary ray generation.

`Sub-pixel jitter sampling`

ARCHITECTURE • MODULE 02

Spatial Acceleration (BVH)

Bounding Volume Hierarchy

Organizes scene geometry into a hierarchical tree of axis-aligned bounding boxes (AABB) to cull unnecessary intersection testing.

`Complexity O(log N) traversal`

Surface Area Heuristic

Builds optimal tree partitions using SAH cost estimation to maximize spatial pruning efficiency in dense polygon environments.

`Optimal tree partitioning`

ARCHITECTURE • MODULE 03

Shading & Material Engine

BRDF & Scattering

Simulates diffuse reflection, metallic specular gloss, and dielectric glass refraction governed by Fresnel equations and Snell's Law.

`Physically based shading (PBR)`

Texture & Color Sampling

Handles image texture mapping, procedural Perlin noise patterns, and gamma correction for precise tone reproduction.

`High dynamic range (HDR)`

ARCHITECTURE • MODULE 04

Parallel Rendering & Export

Multi-Core Scheduling

Distributes pixel scanlines dynamically across all available CPU threads using OpenMP runtime schedulers.

`Near-linear multi-core scaling`

Image Buffer & Export

Accumulates multi-sample anti-aliased pixel color data and exports finalized renders directly to standard PPM format.

`Lossless pixel output`

RESULTS & IMPACT

Project Outcome



12×

Performance boost via BVH spatial acceleration



8+

CPU cores fully saturated with OpenMP threads

BIBLIOGRAPHY

References & Resources

Ray Tracing in One Weekend Series
Peter Shirley, Trevor David Black, and Steve Hollasch (Essential C++ guide).

Physically Based Rendering: From Theory to Implementation
Matt Pharr, Wenzel Jakob, and Greg Humphreys (PBR 4th Edition).

DEMONSTRATION

Working Video / Demo



[Insert working video demo here]

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CONCLUSION

Thank You!

Questions & Discussion